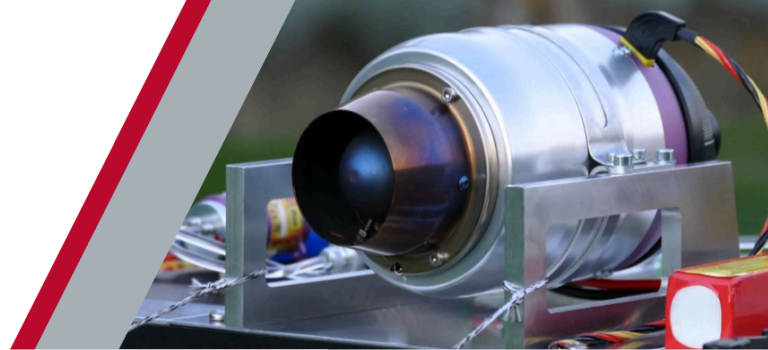




BUCKEYE EXPERIMENTAL AERONAUTICS

# PROPULSION SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Propulsion Subteam is responsible for the design, integration, testing, and validation of all propulsion systems. This includes an electric ducted fan (EDF) system for Prototype 1, turbine engines for Prototypes 2 and 3, and other related subsystems such as fuel systems, engine mounting, thermal protections, and control interfaces.

## Main Responsibilities

- Design and integrate propulsion systems (EDF and turbine)
- Characterize engine performance (thrust, efficiency, thermal)
- Size and develop fuel systems for turbine models
- Design startup, shutdown, and control procedures

## Tools and Equipment

### Software



**Matlab / Simulink**  
Modeling, simulation, controls



**Siemens NX**  
CAD Modeling



**Labview**  
DAQ and Instrumentation



**StarCCM+**  
CFD Software

### Hardware

- JetCat / EDF Engines
- Engine Test Stands
- Pressure Sensors
- Thermocouples
- Fuel System Test Rigs
- Vibration Sensors

## Goals

### Thrust Generation

Deliver reliable, consistent, and predictable thrust across flight regimes

### Safe Engine Operations

Ensure propulsion systems are handled within defined safety limits

### Verifiable Performance and Testing

Validate performance metrics and testing procedures with physical testing

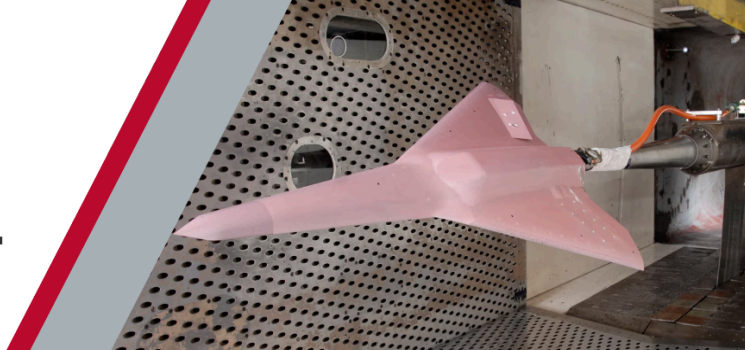
| Prototype                               | Focus                               | Milestones                                                                                                                                                                                                    |
|-----------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | EDF integration and bench testing   | <ul style="list-style-type: none"><li>• EDF system integrated</li><li>• Thrust and performance characterization complete</li><li>• Successful flight testing</li></ul>                                        |
| 2 - Jet Engine<br>(Spring 2027)         | Turbine and fuel system integration | <ul style="list-style-type: none"><li>• Turbine startup/shutdown sequence testing</li><li>• Fuel system safety and function validation</li><li>• Safe flight operations of turbine-powered aircraft</li></ul> |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | High-speed propulsion development   | <ul style="list-style-type: none"><li>• Engine selection for record attempt</li><li>• Flight envelope expansion</li><li>• Engine performance upgrades</li></ul>                                               |



BUCKEYE EXPERIMENTAL AERONAUTICS

# AEROTHERMAL SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Aerothermal Subteam is responsible for understanding and predicting the aerodynamic and thermal environments of the BExA prototype aircraft across all phases of flight. The team develops and validates computational models, conducts wind-tunnel testing, and assess thermal loads to inform design decisions and ensure vehicle performance and safety.

## Main Responsibilities

- Develop and maintain CFD models for aerodynamic performance prediction
- Conduct and validate scale wind-tunnel testing
- Predict thermal loads and perform thermal management analysis
- Support aeroelasticity and fluid-structure interaction simulations

## Tools and Equipment

### Software

-  **StarCCM+**  
CFD Software
-  **Siemens NX**  
CAD Modeling
-  **Labview**  
Wind tunnel DAQ
-  **Paraview**  
Post processing and visualization
-  **Ohio Super Computer**  
Accelerated Sims

### Hardware

- Subsonic Wind Tunnel
- High-Speed Tunnel
- Pressure Scanner/Taps
- Keil Probe Rakes
- High-Speed Computers

## Goals

### Accurate Aerodynamic Predictions

Deliver validated CFD models to predict aerodynamic forces and moments

### Thermal Load Assessment

Quantify surface and internal thermal loads to guide thermal protection and cooling.

### Validation Through Testing

Use wind tunnel and flight data to validate and refine models and guide future designs.

| Prototype                               | Focus                                               | Milestones                                                                                                                                                                                                     |
|-----------------------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | Baseline aerodynamic and stability characterization | <ul style="list-style-type: none"><li>• CFD model workflow</li><li>• Subsonic wind tunnel testing and validation</li><li>• Stability and control derivative estimation</li></ul>                               |
| 2 - Jet Engine<br>(Spring 2027)         | Aerodynamic refinement and thermal analysis         | <ul style="list-style-type: none"><li>• Refined CFD for transonic and supersonic regimes</li><li>• Thermal load mapping and material assessment</li><li>• Scale-model testing in a high-speed tunnel</li></ul> |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | High-speed aeroshell optimization                   | <ul style="list-style-type: none"><li>• Aeroshell optimization for drag reduction at high-speeds</li><li>• Aeroelastic and FSI assessments</li><li>• Coupled Aerothermal and propulsion simulations</li></ul>  |



BUCKEYE EXPERIMENTAL AERONAUTICS

# STRUCTURES SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Structures Subteam is responsible for the design, analysis, manufacturing, and testing of the BExA prototype aircraft primary structure. The team ensures the vehicle can withstand all the aerodynamic, inertial, thermal, and other operational loads across all phases of flight while minimizing weight and maintaining structural integrity, safety, and quality.

## Main Responsibilities

- Design and optimize primary and secondary structures
- Perform structural analysis (FEA) for strength, stiffness, buckling, and vibrational modes
- Develop and validate composite layup schedules and manufacturing processes
- Design mounting schemes for aircraft subsystems

## Tools and Equipment

### Software

**Siemens NX**  
CAD Modeling

**SimCenter**  
Structural FEA

### Hardware

- 3D Printers
- Layup Consumables
- Vacuum Pump
- Tooling Board
- Machine Shop

## Goals

### Structural Integrity

Design a lightweight structure that safely withstands all expected loads

### Manufacturability

Develop repeatable, efficient manufacturing processes for composite structures

### Validation Through Testing

Confirm structural performance through analysis, test, and model correlation

| Prototype                               | Focus                                      | Milestones                                                                                                                                                                                                                          |
|-----------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | Baseline structural design and validation  | <ul style="list-style-type: none"><li>• FEA for static, buckling, and modal performance</li><li>• Manufacturing process development</li><li>• Subsystem mounting and interface designs</li></ul>                                    |
| 2 - Jet Engine<br>(Spring 2027)         | Composite manufacturing process refinement | <ul style="list-style-type: none"><li>• Refined composite structure and layup optimization</li><li>• Thermal and vibration load assessment implementation</li><li>• Structural testing and characterization of components</li></ul> |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | High-speed structural optimization         | <ul style="list-style-type: none"><li>• High-speed aeroelastic and FSI simulations</li><li>• Full airframe structural optimization for weight reduction</li><li>• Structural testing and validation</li></ul>                       |



BUCKEYE EXPERIMENTAL AERONAUTICS

# CONTROLS SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Controls Subteam designs, develops, and validates the dynamic models, control architectures, and autonomous flight systems required to ensure stable, predictable, and repeatable aircraft behavior across all flight regimes. Through simulation and real-world flight testing, the team ensures that control performance remains consistent.

## Main Responsibilities

- Develop linear and nonlinear 6-DOF models
- Develop control laws and set stability targets
- Create abort logic and fault handling
- Simulate and validate performance of varying prototype aircraft designs
- Develop autonomous flight capability
- Extract flight dynamic parameters from test data

## Tools and Equipment

### Software



**MATLAB/ Simulink**  
Modeling, simulation, and control design



**Python**  
Analysis, scripting, autonomous logic



**X-Plane**  
6-DOF Sim and visualization

### Hardware

- Flight Controller
- Servos / Actuators
- Sensor Suite
- DAQ
- RC and Telemetry Links
- Hardware in the Loop

## Goals

### Stability and Control

Develop robust control laws to maintain stability and handle disturbances across all flight conditions

### Autonomous Flight

Enable autonomous flight for precision high-speed passes and navigation

### Safety and Fault Management

Implement fail-safe behaviors, lost-link recovery, and flight envelope protection

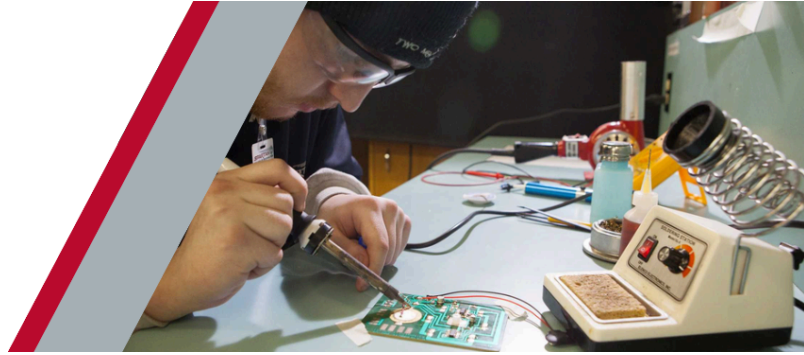
| Prototype                               | Focus                                              | Milestones                                                                                                                                                                                                                     |
|-----------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | Establish baseline models and control architecture | <ul style="list-style-type: none"><li>• Develop 6-DOF model</li><li>• Design baseline autopilot and stability loops</li><li>• HIL simulation and initial flight testing</li></ul>                                              |
| 2 - Jet Engine<br>(Spring 2027)         | Improve processes and expand autonomy              | <ul style="list-style-type: none"><li>• Refine 6-DOF models for transonic conditions</li><li>• Expand autonomy for waypoint following and takeoff</li><li>• Establish live telemetry readout and ground stations</li></ul>     |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | High-speed control optimization                    | <ul style="list-style-type: none"><li>• Full flight-envelope simulation and model validation</li><li>• Fail safe protocols and recovery logic testing</li><li>• Fine-tune control law and autonomous control systems</li></ul> |



BUCKEYE EXPERIMENTAL AERONAUTICS

# AVIONICS SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Avionics Subteam designs, develops, integrates, and validates all electronic systems that support sensing, communication, power distribution, and onboard computation for the aircraft. These systems acquire data, execute flight controls and autonomy functions, manage communications, oversee health and safety monitoring, and ensure reliable operation of BExA prototype aircraft.

## Main Responsibilities

- Define avionics architecture, power distribution, data interfaces, and redundancy strategies
- Integrate and configure flight computers, autopilot, and real-time operating systems
- Select, integrate, and calibrate sensor suites
- Help develop telemetry, RC links, and data protocols for real-time operations.

## Tools and Equipment

### Software



**MATLAB/ Simulink**  
Modeling, simulation, and control design



**Altium**  
PCB Designer



**QGroundControl**  
Autopilot control and mission planning

### Hardware

- Flight Computer
- Soldering Kit
- Telemetry Radio
- Data Logger

## Goals

### Communication Reliability

Maintain strong, low-latency links for telemetry, command, and data transfer

### Robust and Redundant Architecture

Design fault-tolerant systems with redundancy in critical components and communication links

### Power Integrity

Provide regulated power to all systems with monitoring

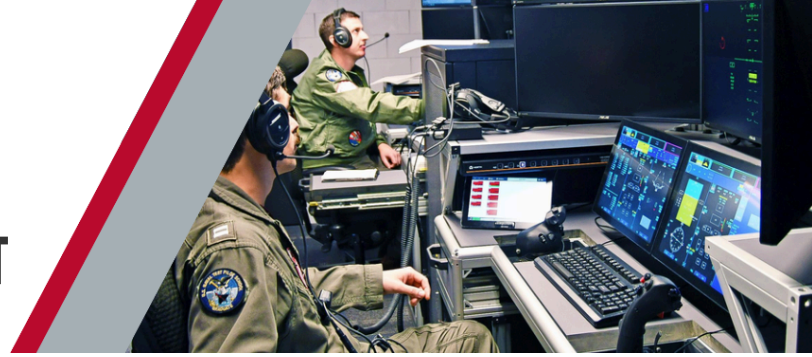
| Prototype                               | Focus                                                  | Milestones                                                                                                                                                                                                                      |
|-----------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | Establish baseline avionics systems and data pipelines | <ul style="list-style-type: none"><li>• Integrate flight computer and core sensor suite</li><li>• Establish telemetry and data logging</li><li>• Validate power systems and wiring harness</li></ul>                            |
| 2 - Jet Engine<br>(Spring 2027)         | Improve reliability and expand capability              | <ul style="list-style-type: none"><li>• Implement advanced health monitoring</li><li>• Increase telemetry ranges and robustness</li><li>• Integrate high-rate data logging</li></ul>                                            |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | Support high-speed autonomy                            | <ul style="list-style-type: none"><li>• Optimize system for low latency and high bandwidth</li><li>• Validate system under high-vibration conditions</li><li>• Support autonomous system development and safety logic</li></ul> |



BUCKEYE EXPERIMENTAL AERONAUTICS

# FLIGHT TEST SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Flight Test Subteam plans, executes, and evaluates all flight operations to ensure safe, controlled, and repeatable testing of BExA prototype aircraft. Responsibilities include test planning, ground system set up, instrumentation, and data collection. The team’s mission is to conduct rigorous, data-driven flight testing while maintaining the highest standards of safety.

## Main Responsibilities

- Develop test plans, objectives, and success criteria for all flight phases and prototypes
- Coordinate range logistics, airspace authorizations, and safety compliance
- Execute pre-flight, launch, flight, and recovery procedures
- Perform risk assessments, enforce safety protocols, and maintain operational safety

## Tools and Equipment

### Software



**Labview**  
Telemetry monitoring and logging



**MATLAB / Simulink**  
Data processing and visualization



**QGroundControl**  
UAV Control and telemetry

### Hardware

- Ground Control Station
- Recovery Parachutes
- Fire Suppression Tools
- DAQ
- Weather Data Station

## Goals

### Safe Operations

Execute all flights safely with no damage to equipment or personnel

### Test Efficiency and Model Validation

Maximize useful data collected per flight through well planned test objectives

### Iterative Advancement

Enable safe, iterative progression towards higher speeds and more complex flight objectives

| Prototype                               | Focus                                                    | Milestones                                                                                                                                                                                                 |
|-----------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | Establish baseline flight operations and data collection | <ul style="list-style-type: none"><li>• Develop preflight, flight, and post flight checklists</li><li>• Collect baseline performance data</li><li>• Perform stable flight in nominal envelope</li></ul>    |
| 2 - Jet Engine<br>(Spring 2027)         | Establish safe turbine testing standards                 | <ul style="list-style-type: none"><li>• Complete propulsion testing safety checklist</li><li>• Complete jet propulsion flight</li><li>• Launch and recovery system testing</li></ul>                       |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | Safe high-speed testing readiness                        | <ul style="list-style-type: none"><li>• Conduct transonic regime flight testing</li><li>• Execute full mission profile testing</li><li>• Validate aircraft in testing and record attempt configs</li></ul> |



BUCKEYE EXPERIMENTAL AERONAUTICS

# BUSINESS SUBTEAM REPORT

2026 - 2027 Year



## Introduction

The Business Subteam manages the financial, sponsorship, and outreach activities that enable the BExA to design, build, test, and achieve its mission. The team develops funding strategies, maintains financial accountability, and builds partnerships that provide resources and visibility for the long-term success of the program.

## Main Responsibilities

- Develop budgets, track expenses, and ensure proper allocation of funds across subteams
- Identify and secure sponsorships, manage external relationships, and organize events
- Manage social media, create promotional content, and represent BExA at events.

## Tools and Equipment

### Software

-  **Adobe Suite**  
Photo and Video Editing
-  **Excel**  
Budgeting and Financials
-  **PowerPoint**  
Presentations
-  **Wix**  
Website Maker

### Hardware

- Information Cards
- Posters
- Banners / Holders
- Vinyl Cutter

## Goals

### Funding Strategy and Accountability

Build and execute multi-year funding strategies to support development and flight testing

### Sponsor Outreach and Relations

Cultivate strong relationships with industry sponsors and maintain interest

### Public Engagement

Promote BExA's mission and accomplishments through consistent communications and social media postings

| Prototype                               | Focus                                            | Milestones                                                                                                                                                                                                                               |
|-----------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 - EDF<br>(Fall 2026)                  | Establish foundations and secure initial support | <ul style="list-style-type: none"><li>• Develop budget and financial tracking system</li><li>• Create sponsorship packages and pitch decks</li><li>• Secure an initial set of sponsors</li></ul>                                         |
| 2 - Jet Engine<br>(Spring 2027)         | Expand sponsorships and increase funding         | <ul style="list-style-type: none"><li>• Expand sponsor networks and engagement methods</li><li>• Submit grant proposals and club funding applications</li><li>• Host outreach events</li></ul>                                           |
| 3 - High-Speed<br>(Fall 27 - Spring 28) | Sustained funding and team visibility            | <ul style="list-style-type: none"><li>• Expand publicity around BExA's mission on social media</li><li>• Establish alumni network and industry partnerships</li><li>• Secure further sponsorships and multi-year funding plans</li></ul> |

